SOPAAN Every step, signed.

An offline-first AI platform that turns a rural student's spoken goals into a verified next step, portable proof of skills, and sponsor funding that follows proof.

Citi Ada Lovelace Hackathon | Reimagining Education and Economic Empowerment for Rural India | Hackathon edition v1.1, 16 October 2026 | Spring Boot + FastAPI, all on free tiers

1. Problem and solution

Radha is in Class 10 in rural Maharashtra. She is good with numbers and sewing machines, has never heard of a polytechnic diploma or an apprenticeship scheme, and her teacher supports 60 students on patchy 3G with low-end phones. The information exists. What is missing is a way to turn it into a step she can **trust**, **complete** and **prove**. Three things break:

What breaks	What it costs	How Sopaan fixes it
Guidance cannot be trusted: scheme rules are scattered, English-heavy and change often; chatbots sound sure and can be wrong.	Wasted trips, missed deadlines, rejected applications.	Reviewed, versioned, sourced rule packs decide eligibility on the device. The AI listens, explains and drafts rules; it never decides.
Skills cannot be proven: tailoring, repair and farm skills have no portable, verifiable evidence.	Good skills earn no recognition or access.	Skill Passport: an authorised teacher or assessor signs a credential; any phone verifies it by QR with no internet.
Funding cannot follow results: sponsors cannot verify who completed what.	Money is spent on activity, not outcomes.	Outcome-linked credits unlock only after a valid credential; capped, counted and reconciled at sync.

Core thesis. AI understands the student. Reviewed rules decide. Signatures prove. Sponsors pay on proof. And it keeps working when the network does not, because **trust lives in the signature, not the channel**: packs, credentials and credit tokens can travel by cloud, QR, WhatsApp file or Bluetooth and are verified identically.

The product in one chain: Discover, Plan, Act, Prove, Release

Link	Student / teacher experience	Record it produces (checkable by the next link)
Discover	Speaks goals in Hindi or Marathi; app repeats back with icons; student confirms each fact	Confirmed profile with confidence and source per field
Plan	3+ diverse options (vocational, technical, academic, enterprise) and an ordered pathway	Pathway built from a prerequisite graph; progress only from events
Act	Next-Step Card: ELIGIBLE / NOT ELIGIBLE (rule) / NEED INFO, documents, deadline, place; Family Voice audio	Eligibility result citing rule id, pack version, source, date
Prove	Teacher signs a skill; employer or college scans a QR	Ed25519 credential; offline verification with registry age shown
Release	Sponsor credit unlocks a training seat or module	Capped, counted token reconciled at sync; duplicates flagged

Lean canvas at a glance

Unique value proposition	An answer a rural student can trust, a skill anyone can verify, and funding that follows proof, in her language and with no internet.
High-level concept	DigiLocker plus a scholarship eligibility checker, built offline-first for villages, with UPI Lite-style capped credits that unlock on proven skills.
Customer segments	Users: rural students Std 9 to graduation and families (free). Distribution: teachers, facilitators, NGOs. Payers: NGOs/CSR, employers, training providers.
Channels	Teachers and NGO coordinators, CSR-funded learning centres, WhatsApp-shared packs, PWA install by link or QR (no app store).
Unfair advantage (being built)	Reviewed sourced rule data, a registry of authorised issuers, auditable outcome records, and offline-by-signature architecture.

How it maps to the judging criteria

Criterion (weight)	Sopaan's answer	Section
Innovation and originality (25%)	One verifiable chain from rule to result to credential to sponsor release; trust in the signature over any channel; AI proposes, rules decide	1, 3
Technical feasibility and architecture (20%)	Modular Spring Boot + FastAPI, deterministic rules engine, signed packs, idempotent sync, tested failure modes, free-tier deployment	4, 5, 6
Usability and scalability (20%)	Voice and icons, Next-Step Card, small bundles, heavy work on device, measured scaling triggers	2, 7
Business value and monetisation (15%)	Free for students; NGO/CSR, employer, provider and facilitator revenue; unit-cost target	7
Customer impact and UX (20%)	From "I do not know what is possible" to a verified, provable next step in her language	1, 8

2. Built to the brief: requirement, response, proof in the demo

Brief requirement	Sopaan response	Demo proof
(a) Local-language voice in/out, simple UI for low literacy	Closed-grammar offline speech recognition (Vosk WASM) per question, icon read-back and confirm, pre-recorded audio clips plus device speech synthesis, text fallback. Hindi first; a language is a data pack (grammar, strings, audio).	Step 1
(b) Low-end Android, offline, low-bandwidth sync	PWA with app shell and core pack under 2 MB, IndexedDB, write-ahead outbox, batched idempotent sync (200 events per call), signed packs movable by file or QR.	Steps 4, 5, 8
(c) Facilitator mode: one teacher, many students	Hub (same backend on the teacher laptop): roster, missing documents, assessments, credential issue, module authoring with revenue share.	Step 6
(d) AI grounded in real data	8 to 10 real schemes encoded as reviewed rules, each with source URL and last-verified date; every claim shown to a student cites rule id and pack version.	Step 2
(e) Easily accessible and extendable	New language, scheme, occupation or issuer is a signed data pack, not a code release. Install by link; no app store.	Pack import, step 5
TR1 Open-source architecture	Open repository and open stack; open data formats for packs and the signed envelope; documented extension points.	Repo
TR2 Meaningful intelligence	LLM and ML used where language is uncertain, with human gates (section 3). Rules, signatures and ledgers handle correctness.	Steps 1, 2
TR3 Multilingual and voice	Speech-to-text offline (closed grammar) and online (provider adapter); text-to-speech via clips and device voices.	Step 1
TR4 Low-bandwidth, offline-first	Local database is the system of record for a session; server is a reconciler; honest freshness badges on every result.	Steps 4, 7
TR5 APIs and interoperability	Versioned OpenAPI REST; public registry and pack endpoints; the signed envelope can be verified by any college, employer or NGO in a few dozen lines. DigiLocker and NSP adapters are planned, not confirmed.	API docs

Out-of-scope reading. The brief excludes villages with no internet at all. Sopaan assumes intermittent connectivity (a student or facilitator gets online sometimes) and treats a zero-connectivity village as a stretch case served by signed files and QR carried by the facilitator.

Team meeting insights, answered

Insight	Where Sopaan answers it
Scalability as evaluation	Heavy work on devices; server load grows with sync events, not taps; stateless workers and measured triggers (section 7).
Secure solution	Ed25519 signatures everywhere, JWT plus device signatures, leased issuer keys with epoch fencing, hash-chained audit, threat model (section 6).
Responsible and ethical AI, controls, no negative implication	AI never decides eligibility, credentials or credits; human review of AI-drafted rules; counterfactual fairness tests; consent and minimal data.
Customer perspective, simplicity	One chain, five links; icon-first screens; Next-Step Card shows exactly what to do next.
Modern engineering, cloud for scale, technology infrastructure	Modular Spring Boot monolith plus FastAPI, event idempotency, containerised, CI that tests and signs packs, clear path from free tier to managed cloud.
AI with a purpose, not just a name	Each AI use is listed with its job, guardrail and offline fallback (section 3).
Include metrics in the submission	Metrics and targets table in section 7.

3. AI that earns its place

The rubric rewards AI-powered personalisation and intelligence, and the meeting asked for purpose over name-sake. Sopaan uses AI where language and uncertainty are the problem, puts a human or a rule behind every consequential output, and keeps a deterministic fallback so nothing breaks offline.

AI use	What it does	Runs	Guardrail / fallback
Voice profiling	Speech to slots (class, category, income band, interests, skills); online, an LLM extracts slots from free speech	Offline grammar STT; online provider + FastAPI /profile:extract	Student confirms every field (human in the loop); icon picker fallback
Rule extraction (the headline AI)	LLM reads scheme documents and proposes structured rules, each criterion quoting a verbatim source span	FastAPI /rules:extract, build time	Output is PROPOSED only; 2 of 3 reviewers co-sign; golden tests must pass before a pack is signed
Career matching and personalisation	Weighted score (interest, skills, cost, distance, prerequisites) then diversity re-rank across clusters so non-STEM and vocational options surface	On device, no network	Guarantee of 3+ clusters when they exist; catalogue is a signed pack
Pathway planner	Prerequisite graph to an ordered plan with documents and deadlines	On device	A step is complete only from a recorded event
Explanation and Family Voice	Turns a rule result into simple local-language text and audio, grounded on the result object only	Templates and pre-recorded clips offline; LLM online	Unsupported-claim check; template fallback
Fairness harness	Counterfactual profiles (change gender, region, language) must give equivalent outcomes unless a rule differs	CI (FastAPI tooling)	Reports pass rate; every unexplained gap is investigated

AI never does this	Because
Decide eligibility; declare a student eligible	A wrong "you qualify" costs a family a trip and a deadline. Reviewed rules can be inspected, tested and corrected; a model score cannot.
Issue or validate credentials; release credits	These are signature, lease, cap and counter checks in deterministic code.
Ship a rule without human approval	Documents are untrusted input (prompt-injection risk); reviewers sign the pack.

Orchestration. The five bounded agents (Profile, Career Discovery, Pathway, Opportunity Matcher, Explanation) are nodes of one DAG with a human-confirmation node after Profile. The same DAG has two executors: a deterministic state machine on the device (no LLM) and the FastAPI orchestrator online (LLM allowed for Profile and Explanation). Tool arguments are schema-validated, calls have timeouts, and personal data is stripped before any external provider call.

Responsible AI and privacy in practice

- Pseudonymous student IDs; the name roster stays with the facilitator; no Aadhaar by default; raw audio stays on the device unless the student opts in online.
- Versioned consent and guardian consent for minors (review Digital Personal Data Protection Act 2023 child-data provisions before any pilot). Sponsors see aggregates only.
- Every result shows its source, rule version and age; stale or cached results are labelled, never presented as current.
- No public rankings; talent indicators are visible and correctable by the student and never used to deny access.

Facilitator mode and monetisation hooks

The hub gives one teacher a cohort dashboard (who is missing documents, who is eligible for what, who completed which module), lets them issue credentials under a leased issuer key, and author training modules that they can offer free, per module or by subscription, keeping the majority of the revenue.

4. Architecture (high level)

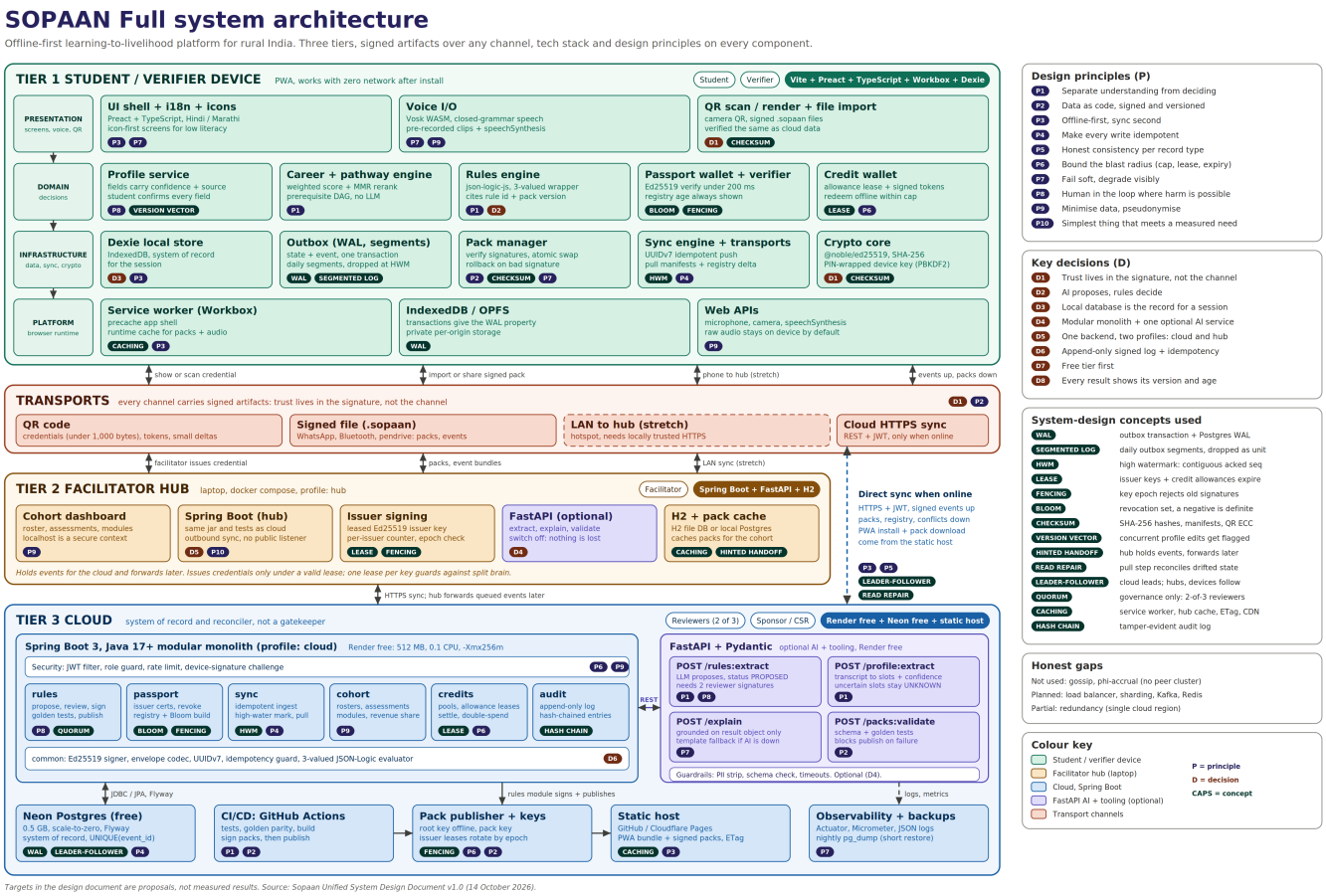


Figure 1. Full system architecture: three tiers, transports, and the design principles (P), decisions (D) and system-design concepts applied to each component.

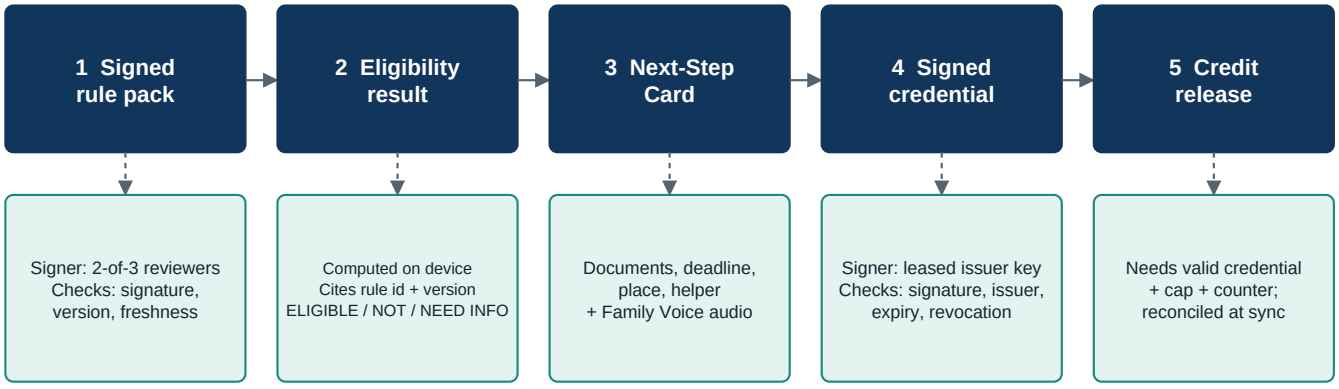


Figure 2. The trust chain: each link produces a record the next link can verify.

Layer	Choice and role
Student app	Vite + Preact + TypeScript PWA, Workbox service worker, Dexie/IndexedDB; json-logic-js 3-valued rules; @noble/ed25519; Vosk WASM voice
Backend core	Spring Boot 3 modular monolith (rules, passport, sync, cohort, credits, audit); profiles "cloud" and "hub"; Flyway; JWT + signed events
AI and tooling	FastAPI + Pydantic: rule extraction, profile extraction, explanation, pack validation, fairness harness (all optional at runtime)
Data and delivery	Neon Postgres (cloud), H2 (hub); GitHub Actions builds, tests, signs packs; static host for PWA and packs

5. Low-level design

5.1 Wire formats: one signed envelope for everything

```
Envelope = { "v":1, "kid":"<key id>", "typ":"pack|cred|registry|event|token|allowance",
             "payload":"<base64url bytes>", "sig":"<Ed25519 over the exact payload bytes>" }   // no JSON canonicalisation needed

Rule (inside a pack):  { "id":"PMS-OBC-MH", "source":{"url":"https://...","verified":"2026-10-10"},
  "criteria":{"all":[ {"id":"c1","logic":{"<=":[{"var":"family_income"},250000]},"needs":["family_income"]},
                      {"id":"c2","logic":{"in":[{"var":"category"},["OBC","SC","ST"]]},"needs":["category"]} ]},
  "documents":["income_certificate"], "deadline":"2026-12-31" }  + golden cases {profile, expect: ELIGIBLE | NOT:c1 | NEED:[fields]}

Credential: { "id":uuid, "iss":kid, "epoch":3, "sub":uuid, "skill":"TAILOR.L2", "ev":sha256, "iat":..., "exp":... }  (~350 B + 64 B sig, one Q
```

5.2 Core algorithms

```
ELIGIBILITY (3-valued):  leaf -> FALSE if logic false;  UNKNOWN if a needed field is missing;  TRUE otherwise
  ALL: any FALSE -> NOT_ELIGIBLE(first failing rule id);  else any UNKNOWN -> NEED_INFO(fields, documents);  else ELIGIBLE

VERIFY OFFLINE:  issuer cert known?  -> signature valid?  -> epoch >= cert.epoch and iat inside lease?  -> not expired?
  -> Bloom(cred.id) negative = not revoked as of registry date;  positive = POSSIBLY_REVOKED (check exact list / go online)
  always display registry age

SYNC PUSH (server):  for each event (<=200): verify device signature + schema; INSERT ... ON CONFLICT (event_id) DO NOTHING;
  apply projections in the same transaction;  return acked_upto = contiguous high-water mark; client prunes outbox segments <= hwm

CREDIT:  mint requires valid credential matching the pool milestone, unexpired allowance, ctr == last+1, spent+amt <= cap
  settle: UNIQUE(allow_id, ctr); token not already settled elsewhere; violation -> FLAGGED (loss bounded by cap)
```

5.3 Components and data model

Area	Design
Local DB (Dexie)	profile (value + version vector), events (WAL outbox in daily segments), packs, registry, credentials, allowance, tokens, progress, keys (PIN-wrapped)
Spring modules	rules (pack lifecycle, golden tests, signer), passport (issuers, credentials, revocation, Bloom), sync (ingest, high-water mark), cohort, credits (pools, leases, settlement), audit (hash chain)
Core tables	student, device, event(event_id PK, UNIQUE(device_id, seq)), rule_pack, issuer_cert, credential, revocation, credit_pool, credit_allowance, credit_token, consent, audit_log
REST (OpenAPI)	POST /v1/sync/push, GET /v1/sync/pull, GET /v1/packs/{name}/{version}, GET /v1/registry/snapshot, POST /v1/credentials, POST /v1/admin/rules/{id}:approve, POST /v1/credits/settle
FastAPI	POST /v1/rules:extract, /v1/profile:extract, /v1/explain, /v1/packs:validate; CLI eval/fairness
Concepts applied	WAL, segmented log, high-water mark, lease, fencing (key epoch), Bloom filter, checksums, version vectors, hinted handoff, read repair, leader-follower, CAP/PACELC (availability for cached guidance, consistency for consent, ledger, revocation)

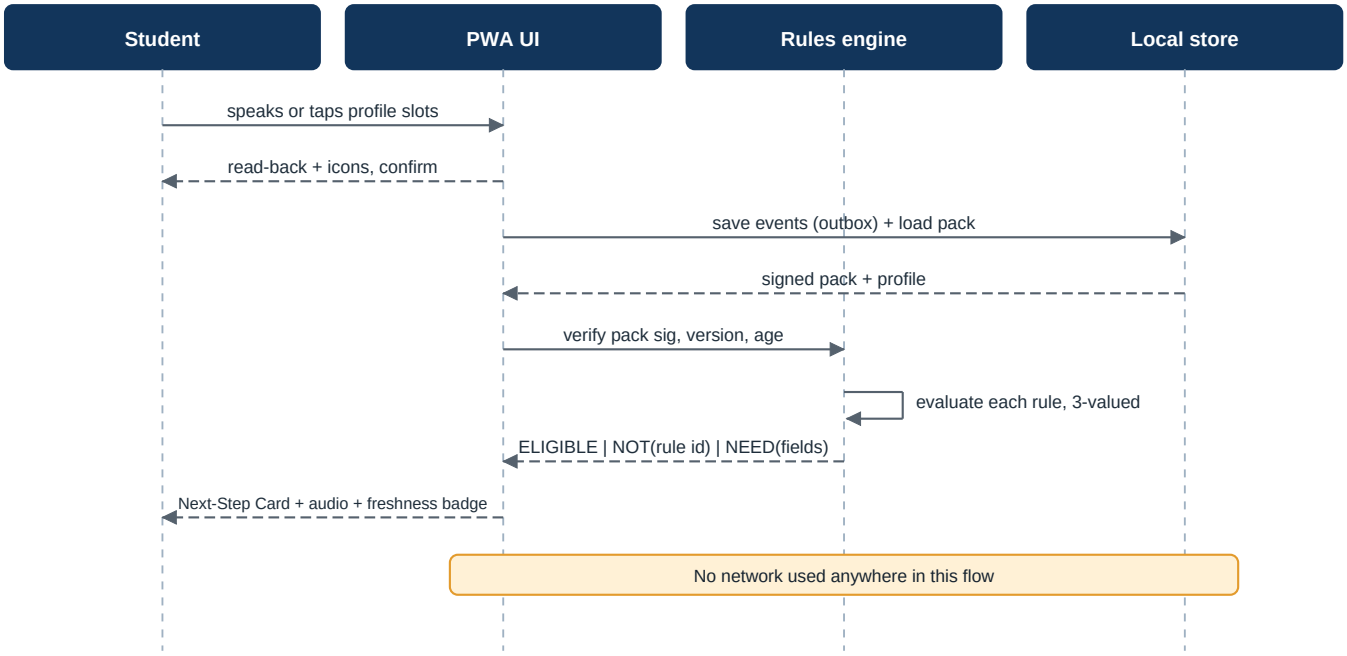


Figure 3. Sequence: voice profile to Next-Step Card with no network.

6. Key flows, failure handling and security

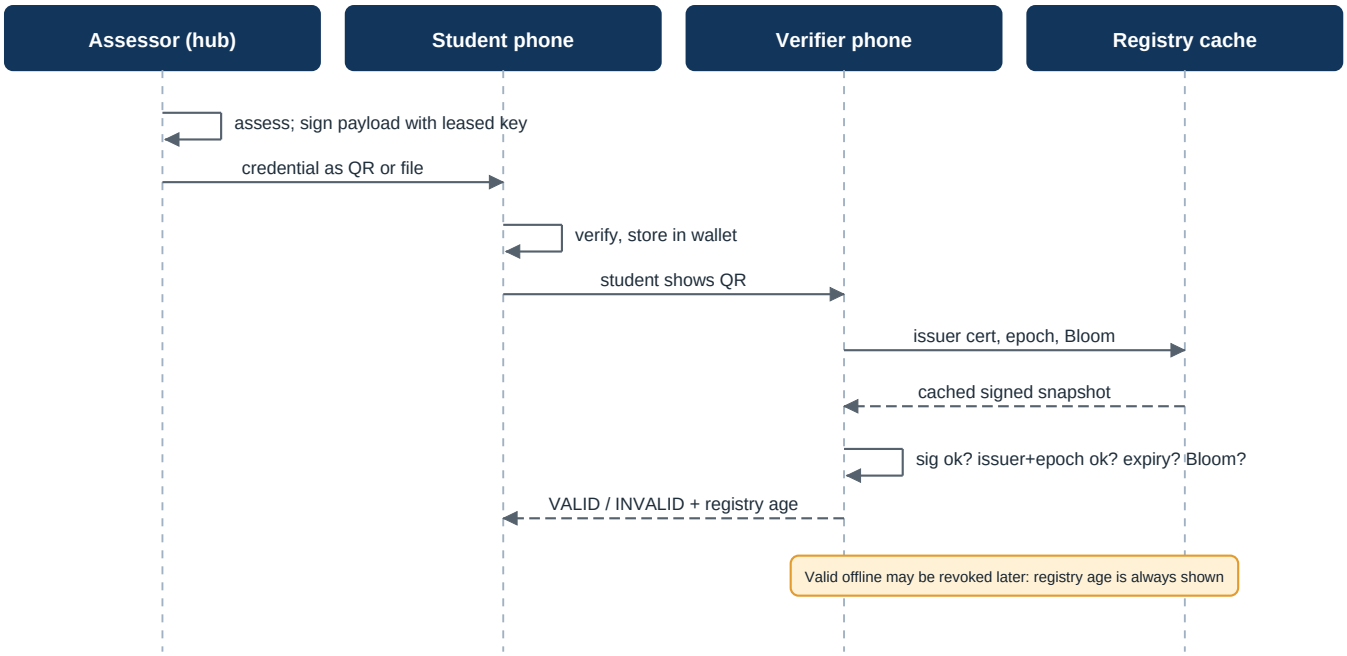


Figure 4. Sequence: issue a credential under a leased key and verify it offline against a cached registry.

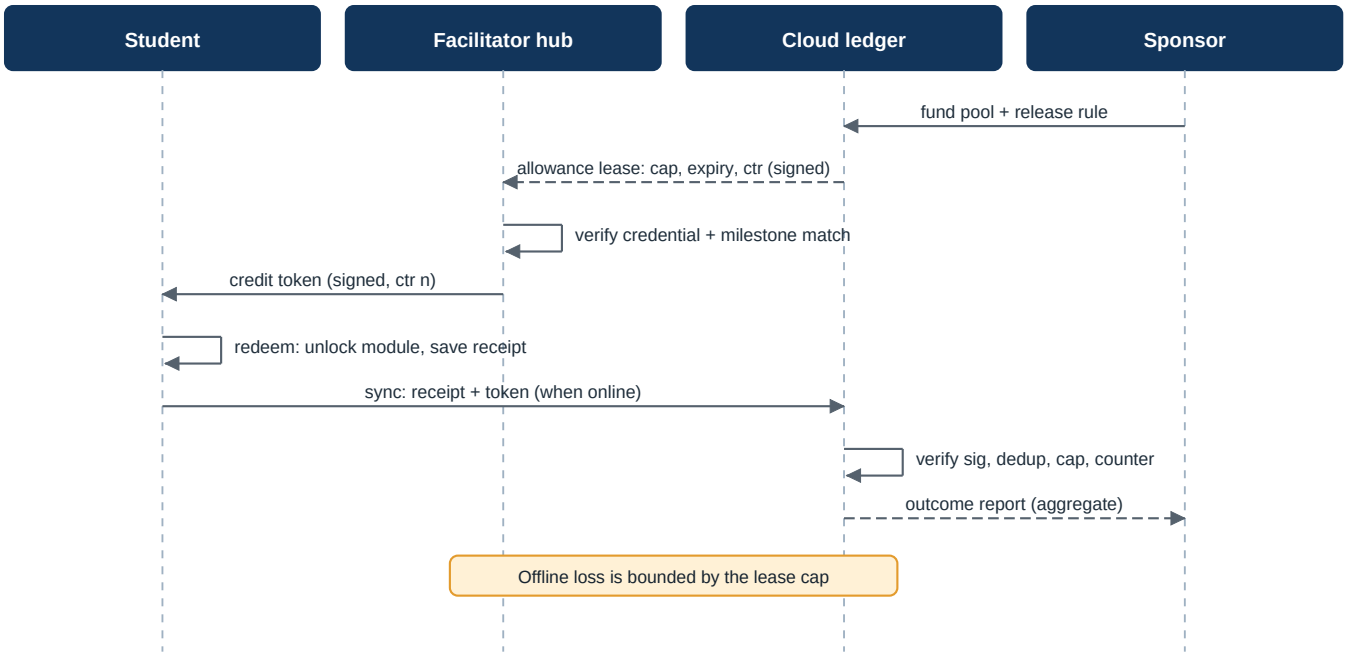


Figure 5. Sequence: sponsor funding to offline redemption and reconciliation. Offline loss is bounded by the lease cap.

Failure or threat	What happens	Mitigation
No network	All offline-column flows work; new actions show "pending sync"	WAL outbox, retry with backoff
Pack past expiry; old or forged pack	Stale badge; "update rejected, still using version X"	Warn never block; atomic swap only after full verification
Tampered or forged credential	INVALID: bad signature	Ed25519, issuer registry, epoch fencing, pinned root key
Duplicate event after retry	Nothing visible	UNIQUE(event_id); duplicate is success
Credit used on two devices	Flagged at settlement	Cap bounds loss; audit entry
Prompt injection via scheme documents; AI writes rules	Rule stays PROPOSED	Documents are data; schema + citation check; 2-of-3 reviewers
Facilitator sees another cohort; sync flooding	Denied or throttled	Server-side ownership checks, role guard, batch and rate limits
AI provider down	Template explanation, icon forms	Fallback chain; AI is optional

A valid signature proves who signed and that nothing changed, not that the assessment was fair, so issuers are authorised, scoped by skill, leased with expiry and held to rubrics.

7. Scale, free-tier deployment, business model and metrics

Free-tier fact (checked 14 Oct 2026; re-check before submitting)	Consequence
Render free web service: 512 MB, 0.1 CPU, sleeps after 15 idle minutes, about a minute to wake	Offline-first sync hides cold starts; lean JVM; demo runs on the local hub
Neon free Postgres: 0.5 GB, 100 compute-hours per month, no card	About 50 MB per month for a 1,000-student pilot (estimate), so roughly 10 months before compaction or upgrade
Oracle Always Free Arm allowance reduced in 2026 and often unavailable; Hugging Face compute Spaces need a paid plan	Not relied on
Sarvam speech: credit and price pages disagree on free credit (Rs 100 vs Rs 1,000)	Used only to pre-generate audio and as an online option

Scalability

Heavy, frequent work (rule evaluation, verification, plans) runs on devices; the cloud handles sync, registry and optional AI, so load grows with events, not taps. Scale only what is measured: stateless workers and batching first, then Redis and CDN for packs, a log such as Kafka for ingest, extraction of a module into its own service, and sharding by district. A 60-student class sync is about 1.5 MB in total.

Business model (illustrative planning assumptions, to validate in a pilot)

Payer	Pays for	Illustrative
Students and families	Nothing; guidance and Skill Passport are free	Free
NGOs, CSR programmes	Cohort tools, verified outcomes, aggregate impact reports	About Rs 60 per student per year
Employers, apprenticeship providers	Verified candidates, credential verification	About Rs 500 to 1,000 per enrolled candidate
Training providers, institutions	Listings, issuer onboarding	To validate
Facilitators earn	Module sales and commissions	Keep the majority; platform share 10 to 15%

Cost target: under Rs 3 per student per month in variable AI cost, because offline-first and pre-generated audio avoid most cloud turns. Credits are simulated, non-cash entitlements in the prototype; Sopaan is not a lender, scholarship authority or payment provider, and a credential never acts as a credit score. Citi alignment is potential, not an existing partnership.

Metrics for the idea submission (targets, not results)

Metric	Target	Metric	Target
Must flows complete in airplane mode	100%	Eligibility agreement on reviewed golden set	100% (state set size)
Results citing rule id, version, source	100%	Valid events synced; zero duplicate effects	>= 99%; 0
Offline slot word error rate (closed vocabulary)	< 15%	Recommendation Recall@5; non-STEM options shown	>= 0.80; >= 3
Cold start on 2 GB Android; offline verify time	< 5 s; < 200 ms	Students completing the primary journey	>= 80%
Credential size	<= 1,000 B	Variable cost per student per month	< Rs 3

8. Demo, scope, risks and next steps

Live demo: Radha's story (8 steps)

#	Step	What the judges see
1	Discover	Hindi voice profile; the app reads back with icons; Radha confirms
2	Verified eligibility	Scholarship result with the deciding rule, source, version and missing documents; show the AI-extracted rule and its reviewer approval
3	Next step	Next-Step Card and Family Voice audio
4	Airplane mode	Same check runs from the cached pack
5	Pack by WhatsApp	A newer signed pack arrives as a file offline; the app verifies it and switches versions
6	Prove	Facilitator issues a credential; a second phone verifies it from a QR
7	Failure cases	Tampered credential rejected; stale-registry warning shown
8	Money follows proof	Simulated credit unlocks only after a valid credential; reconnect, sync, replay an event, no duplicate

Scope tiers

Tier	Items
MUST	Offline PWA; 8 to 10 real reviewed scheme rules with sources; shared golden tests (TypeScript and Java); signed packs and file import; credential issue and QR verify; tamper and stale cases; idempotent sync; closed-grammar voice in one language; Next-Step Card; facilitator dashboard; simulated credits
STRETCH	FastAPI rule extraction with reviewer queue (needed to show AI in the demo, so schedule it early); fairness CI; animated-QR transfer; LAN hub sync with trusted HTTPS; second language
PLANNED (label as such)	Real payments or UPI payouts, lender partnership, DigiLocker/NSP/employer integrations, Kafka, Redis, vector search

Top risks and mitigations

Risk	Mitigation
Scope too large	Strict tiers; cut order: second language, LAN sync, then extra credit features
AI looks optional to judges	Demo rule extraction with citations and reviewer approval, conversational profiling online, grounded explanations
Offline Hindi recognition unproven	Closed grammars, icon fallback, report the measured word error rate honestly
Scheme rules wrong or stale	Source URL and date per rule, two reviewers, pack expiry, report rule correctness separately
Free-tier terms change	Re-verify in submission week; docker-compose fallback; portable stack
PWA install needs a secure context	One online install or sideloaded APK; documented limit; LAN sync is a stretch

Before you submit

- Fill the Lean Canvas columns A to K (team details); the content columns are already filled in **Lean_Canvas-Sopaan.xlsx**.
- Replace evidence placeholders: one verified statistic (for example UDISE+ with year) and 5 to 10 consented interviews with students or teachers.
- Measure offline Hindi slot recognition on a small recording set; pick one language and one state.
- Rehearse the 8 steps on real devices in airplane mode, including the tampered credential.
- Check the name Sopaan (trademark, domain, app stores); describe Citi, government, school and employer relationships as prospective until confirmed.
- Keep targets and measured results separate on every slide and in every spoken claim.